Mathematical Epidemiology: Survival Analysis of 1,183 Erdős Problems

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Abstract

We present, to our knowledge, the first application of survival analysis to the resolution dynamics of a mathematical problem corpus. Treating 1,183 Erdős problems as subjects in a right-censored time-to-event study—with problem index serving as an ordinal time proxy and resolution as the event of interest—we apply Kaplan–Meier estimation, Cox proportional hazards regression, competing risks analysis, and Weibull accelerated failure time modeling. Our principal findings are: (i) prize problems resolve significantly faster (HR = 2.60, 95% CI: 1.85–3.67, $p < 0.001$); (ii) formalized problems exhibit 45% lower hazard of resolution (HR = 0.55, 95% CI: 0.43–0.71, $p < 0.001$), consistent with a selection effect rather than a causal mechanism; (iii) the overall proved-to-disproved ratio is 2.63:1, but varies from 8.5:1 in combinatorics to 1.3:1 in set theory; and (iv) a Weibull AFT model with shape parameter $\hat{\rho} = 1.61$ indicates decreasing marginal hazard, with prize attachment accelerating resolution by a factor of 0.60 ($p < 0.001$). The Cox model achieves a concordance index of $C = 0.60$ with a significant global test ($\chi^2 = 65.4$, df = 7, $p < 10^{-10}$).

1 Introduction

The Erdős problem corpus—1,183 conjectures, questions, and challenges posed by Paul Erdős (1913–1996) and catalogued by Tao et al. [1]—provides an opportunity to study the *dynamics* of mathematical progress through a quantitative lens. Each problem can be viewed as a patient entering a longitudinal study at the time of its formulation, with resolution (proof, disproof, or other definitive answer) constituting the event of interest. Problems that remain open are right-censored.

While bibliometric approaches to mathematical productivity are established [6, 7], survival-analytic methods have not previously been applied to a structured database of open problems. The Erdős corpus is particularly suited: problems span six decades, carry metadata (tags, prizes, OEIS references, formalization status), and exhibit a natural ordering by problem number that serves as an ordinal time proxy. We ask: which covariates predict faster resolution, how does the hazard change across the corpus, and how do the competing risks of proof versus disproof vary by subfield?

Notation. Problem numbers $i \in \{1, \ldots, 1183\}$ serve as the time variable T. The event indicator is $\delta_i = 1$ if problem i has been resolved and $\delta_i = 0$ otherwise (right-censored).

2 Data and Methods

2.1 Dataset

The Erdős Problems database [1] records $n = 1{,}183$ problems with the following fields: problem number, status (open, proved, disproved, solved, falsifiable, verifiable, decidable, independent, or

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Lean-verified variants), tags (mathematical subfields), prize amount (USD or GBP), OEIS sequence references, and formalization status. We restrict the primary analysis to problems with a definitive binary outcome:

$$n_{\text{events}} = 483 \text{ (resolved)}, \quad n_{\text{censored}} = 650 \text{ (open)}, \quad n_{\text{total}} = 1{,}133.$$

The remaining 50 observations with intermediate statuses (falsifiable, verifiable, decidable, independent) are excluded from the main survival models but included in the competing risks analysis as a third event type.

Covariates are summarized in Table 1.

Table 1: Covariate definitions and descriptive statistics ($n = 1{,}133$).

Covariate	Type	Prevalence/Mean	Events (%)
Prize (> \$0)	Binary	8.8% (100/1133)	38.0%
Formalized	Binary	33.4% (378/1133)	21.7%
OEIS sequences	Binary	25.2% (286/1133)	26.2%
Number of tags	Count	$\bar{x} = 1.63$	—
Tag solve rate	Continuous $[0, 1]$	$\bar{x} = 0.40$	—
Number theory	Binary	47.7%	37.2%
Graph theory	Binary	22.3%	49.0%
Ramsey theory	Binary	9.3%	40.9%

2.2 Statistical Methods

Kaplan–Meier estimation. We estimate the survival function $\hat{S}(t)$ using the product-limit estimator [2], with pointwise 95% confidence intervals via Greenwood's formula. Stratified curves are compared using the log-rank test.

Cox proportional hazards regression. The hazard for problem i is modeled as

$$h(t \mid \mathbf{x}_i) = h_0(t) \exp(\boldsymbol{\beta}^\top \mathbf{x}_i),$$

where $h_0(t)$ is the unspecified baseline hazard and $\mathbf{x}_i$ is the covariate vector [3]. Ties are handled via the Breslow approximation. Model adequacy is assessed by the concordance index C, partial AIC, and the proportional hazards assumption is tested via scaled Schoenfeld residuals [4].

Accelerated failure time models. We fit parametric AFT models of the form $\log T_i = \boldsymbol{\mu}^\top \mathbf{x}_i + \sigma W_i$, where W_i follows a standard extreme value (Weibull), normal (log-normal), or logistic (log-logistic) distribution. Model selection is by AIC.

Competing risks. Resolution events are decomposed into proof ($k = 1$) and disproof ($k = 2$) as competing risks. We report cause-specific cumulative incidence functions $\text{CIF}_k(t) = P(T \leq t, \text{cause} = k)$ and proved-to-disproved ratios stratified by tag.

3 Results

3.1 Kaplan–Meier Estimates

The overall Kaplan–Meier curve yields an estimated median survival of $\tilde{t} = 946$ (i.e., the survival function crosses 0.50 at problem number 946). The interquartile range is $[\hat{t}_{0.25}, \hat{t}_{0.75}] = [570, 1134]$. At selected timepoints: $\hat{S}(250) = 0.899$, $\hat{S}(500) = 0.782$, $\hat{S}(750) = 0.654$, $\hat{S}(1000) = 0.451$.

Stratified Kaplan–Meier analysis reveals significant differences by covariate group (Table 2).

Table 2: Log-rank tests for stratified Kaplan–Meier curves.

Stratum	$\tilde{t}_1$	$\tilde{t}_0$	χ^2	p-value
Prize (yes/no)	703	965	24.29	< 0.001
Formalized (yes/no)	1102	900	24.22	< 0.001
OEIS (yes/no)	—	—	10.12	0.001
Graph theory (yes/no)	842	—	2.86	0.091
Ramsey theory (yes/no)	894	—	0.01	0.906

$\tilde{t}_1$, $\tilde{t}_0$: median survival for the indicated and reference group, respectively. "—" indicates the median was not reached or is omitted for brevity.

Prize problems have a median survival of 703, compared to 965 for non-prize problems—a reduction of 262 ordinal units ($\chi^2_{LR} = 24.29$, $p < 0.001$). Conversely, formalized problems exhibit a *higher* median survival of 1102 versus 900, which we discuss in Section 4 as a selection effect.

3.2 Cox Proportional Hazards Regression

We fit a Cox model with structural covariates and subfield indicators (Table 3).

Table 3: Cox proportional hazards model with tag indicators (n = 1,133, events = 483).

Covariate	$\hat{\beta}$	HR	95% CI	p
Prize	0.96	2.60	$(1.85, 3.67)$	< 0.001
Formalized	-0.59	0.55	$(0.43, 0.71)$	< 0.001
OEIS	-0.47	0.63	$(0.48, 0.82)$	< 0.001
Number theory	0.36	1.44	$(1.13, 1.84)$	0.004
Graph theory	0.18	1.20	$(0.93, 1.55)$	0.171
Ramsey theory	-0.18	0.83	$(0.60, 1.16)$	0.284
Analysis	-0.12	0.89	$(0.62, 1.26)$	0.503
$C = 0.60$; partial AIC = 5838.0; LR $\chi^2 = 65.4$, df = 7, $p < 10^{-10}$				

Three covariates achieve significance at $\alpha = 0.01$: prize attachment (HR = 2.60; prize problems have 2.6× the instantaneous hazard of resolution), formalization (HR = 0.55; 45% reduction), and OEIS linkage (HR = 0.63; 37% reduction). Among tag indicators, only number theory is significant (HR = 1.44, $p = 0.004$), indicating that number theory problems resolve faster than the baseline after controlling for other covariates—a finding that may reflect the larger community working in number theory rather than intrinsic tractability.

Ramsey theory shows a non-significant tendency toward lower hazard (HR = 0.83, $p = 0.284$), though the multivariate log-rank test across five tag groups approaches significance ($\chi^2 = 8.63$, df = 4, $p = 0.071$).

The global concordance index of $C = 0.60$ indicates modest but real discriminative ability—consistent with the expectation that problem-level metadata captures only a fraction of the variance in resolution difficulty, the remainder being driven by the (unobserved) intrinsic mathematical content.

3.3 Competing Risks

Decomposing resolution events into proof ($k = 1$) and disproof ($k = 2$), the overall proved-to-disproved ratio is 2.63:1 (305 proofs, 116 disproofs), with 62 additional events classified as "solved" without further specification. This ratio varies substantially across subfields (Table 4).

Table 4: Competing risks decomposition by mathematical subfield.

Subfield	n	Proved	Disproved	P : D ratio
Overall	1133	305	116	2.6:1
Combinatorics	46	17	2	8.5:1
Ramsey theory	105	31	8	3.9:1
Analysis	74	29	9	3.2:1
Number theory	541	133	46	2.9:1
Geometry	101	25	9	2.8:1
Graph theory	253	74	35	2.1:1
Set theory	31	4	3	1.3:1

The P : D ratio ranges from 8.5:1 in combinatorics to 1.3:1 in set theory. Graph theory exhibits an elevated disproof rate (2.1:1), consistent with the availability of explicit counterexample constructions.

3.4 Accelerated Failure Time Models

We fit Weibull (AIC = 7793.7), log-logistic (7854.1), and log-normal (7930.9) AFT models; the Weibull provides the best fit (Table 5). The shape parameter $\hat{\rho} = 1.61$ (95% CI: 1.49–1.74) exceeds 1, indicating *decreasing* marginal hazard—problems surviving to higher indices have progressively lower resolution rates.

Table 5: Weibull AFT model estimates ($n = 1{,}133$, events = 483).

Covariate	$\hat{\beta}$	AF $= e^{\hat{\beta}}$	95% CI	p
Prize	-0.51	0.60	$(0.49, 0.74)$	< 0.001
Formalized	0.32	1.38	$(1.19, 1.61)$	< 0.001
OEIS	0.19	1.21	$(1.03, 1.41)$	0.020
n_{tags}	-0.04	0.96	$(0.87, 1.05)$	0.363
Shape $\hat{\rho} = 1.61$ (95% CI: 1.49–1.74); AIC = 7793.7; $C = 0.58$				

Acceleration factors AF < 1 indicate faster resolution (shorter "survival"); AF > 1 indicates slower resolution.

Prize attachment yields AF = 0.60 (resolution in 60% of the time required by non-prize problems); formalization yields AF = 1.38 (38% longer survival), reflecting the selection effect discussed below.

4 Discussion

The formalization paradox. Our most counterintuitive finding is that formalized problems resolve *more slowly* (HR = 0.55, AF = 1.38). This is not a causal effect; rather, it is a classic *selection bias*. Problems chosen for formalization in Lean or Coq tend to be famous and difficult—precisely those that resist resolution. The formalization indicator acts as a proxy for unobserved difficulty: only 21.7% of formalized problems are resolved, versus 50.1% of non-formalized problems.

The prize effect and Simpson's paradox. The prize hazard ratio of HR = 2.60 admits two interpretations: incentive effects attracting effort [9], and more precise problem statements facilitating targeted attacks. Notably, the raw solve rate for prize problems (38.0%) is *lower* than for non-prize problems (41.2%)—the Cox model reverses this after controlling for formalization and OEIS status, which confound the association. This is Simpson's paradox in survival data.

Competing risks and epistemic calibration. The 2.63:1 proved-to-disproved ratio quantifies Erdős's *epistemic calibration*: his conjectures were correct $\sim 70\%$ of the time among resolved problems. In combinatorics (8.5:1) his intuition was nearly infallible; in set theory (1.3:1) the near-parity suggests genuine uncertainty, consistent with independence phenomena prevalent in that domain.

Limitations. (i) Problem number is an ordinal proxy for time, not a metric variable; the mapping from index to calendar time is nonlinear. (ii) The database lacks true resolution dates, preventing calendar-time survival analysis. (iii) Covariates capture metadata rather than mathematical content; the moderate concordance ($C = 0.58$–0.60) reflects this inherent limitation. Despite these constraints, the survival-analytic framework reveals structure that descriptive statistics alone do not: Simpson's paradox in the prize effect, the selection mechanism underlying the formalization paradox, and field-dependent asymmetry in competing risks.

Implications. As AI-assisted theorem proving accelerates [10, 11], the resolution hazard may increase. Our Weibull baseline provides a benchmark: if the post-2025 hazard exceeds the predicted rate, this constitutes measurable evidence of AI impact on open mathematical problems.

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